

The Editor-in-Chief
Journal of Information Security and Applications
Elsevier

Dear Editor-in-Chief,

Following transfer through Elsevier's journal transfer process, we respectfully submit our manuscript, "**NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction**," for consideration as an Original Research Article in the *Journal of Information Security and Applications* (JISA).

Relevance to JISA. NARCIS is a practical multimedia-security system for transmitting protected messages through sequences of unchanged natural images selected from a shared index. The work connects information-security mechanisms with an empirically validated application: AES-GCM protects payloads and session metadata, associated data binds sender and receiver identities, keyed symbol assignment limits public label exposure, monotonic sequence checking rejects replay, and Reed–Solomon coding supports exact authenticated recovery after image-channel distortions. It therefore directly addresses JISA's interests in authentication, cryptographic protection in systems and applications, and multimedia security.

- **Applied security protocol:** confidentiality, authenticity, identity binding, replay control, framing, error correction, and cover selection are integrated and tested in one end-to-end implementation.
- **Operational robustness:** a codebook is accepted only when every protected coded message can be realised by the finite, attack-qualified image index, rather than on nominal descriptor capacity alone.
- **Adversarial evaluation:** calibration and holdout distortions are separated, while selection leakage is measured with statistical, ensemble, and learned detectors.

Technical Contribution and Evidence. Across five deterministic partitions each of BOSSBase and RGB Caltech-101, all 1,575 clean, calibration, and holdout message-condition trials recovered the authenticated plaintext. Mean symbol accuracies were 98.74% and 98.59%. A matched BOSSBase ablation fell from 210/210 to 142/210 recoveries without qualification. At a 16-symbol operating point, net rate increased from 0.184 to 1.213 bits/cover for 8- to 128-byte payloads. The paper also reports a weak Caltech-101 selection signal (repeated CNN AUC 0.533, 95% CI [0.517, 0.548]) and explicitly limits its security conclusions to the tested datasets, partitions, attacks, threat model, and detector classes.

Positioning Within the Journal's Criteria. The manuscript does not propose a new cryptographic primitive or rely on purely theoretical security arguments. It applies established cryptography within a reproducible communication system and evaluates complete-message security and recovery. Nor is it a routine application of an off-the-shelf AI model to a standard security dataset: the learned descriptor is one component of a new protocol-level design whose principal contributions are attack-qualified finite-index feasibility, protected-message-aware operating point selection, and end-to-end authenticated validation. The implementation, tests, fixed seeds, experimental evidence, and manuscript sources are publicly available.

Mandatory Author Declarations.

- **Originality/exclusivity:** The manuscript is original, unpublished, and not under consideration elsewhere. It is being submitted to JISA through Elsevier's transfer process and has not previously been submitted to or rejected by JISA.
- **Author approval:** All authors approved the manuscript, order, contributions, and submission.
- **Interests/funding:** No competing interests or external funding are declared.
- **Ethics/reproducibility:** No humans, animals, or personal data are involved. Code, seeds, evidence, tests, and sources are available at github.com/EkodeckStephane/NARCIS.

We believe NARCIS will interest JISA readers seeking technically detailed, practice-oriented security solutions that combine cryptographic safeguards, multimedia communication, robustness testing, and explicit analysis of residual detectability. Thank you for considering this manuscript.

Yours sincerely,

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